Technical
Webinar Series on Climate Data Analysis in SSA
Part 1

Cascade Tuholske
Montana State University
Sep 2, 2025

Agenda

- 3:00 – 3:05pm (5mins) Welcome and rules of engagement Hyacinth Edeh
- 3:05 – 3:10 pm (5 mins) Opening Remarks IFPRI Country Program Leader, Dr. Oliver Kirui
- 3:10 – 3:45pm (35mins) Webinar presentation, Prof. Cascade Tuholske
  - NASA Project Overview
  - Introduction to **observational** climate data for health analysis
- 3:45 – 3:50pm (5 mins) Short Break All
- 3:50 – 4:25pm (35 mins) Webinar presentation continues, Prof. Cascade Tuholske
  - Introduction to **observational** climate data for health analysis (continued)
  - Q & A (15 minutes)
- 4:25pm – 4:30pm (5 mins) Closing and next step Dr. Oliver Kirui and Prof. Cascade Tuholske

A reminder …


CALL FOR APPLICATIONS
Technical Webinar Series on Climate Data Analysis in SSA
Tuesday September 2, 2025
8-9:30 am MST / 10-11:30 am EST / 3-4:30 pm GMT +1
Tuesday October 7, 2025
8-9:30 am MST / 10-11:30 am EST / 3-4:30 pm GMT +1
Tuesday November 4, 2025
8-9:30 am MST / 10-11:30 am EST / 3-4:30 pm GMT +1
Tuesday December 2, 2025
8-9:30 am MST / 10-11:30 am EST / 3-4:30 pm GMT +1
Tuesday January 20, 2026
8-9:30 am MST / 10-11:30 am EST / 3-4:30 pm GMT +1
Tuesday February 3, 2026
8-9:30 am MST / 10-11:30 am EST / 3-4:30 pm GMT +1
INTERNATIONAL FOOD POLICY RESEARCH INSTITUTE
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The Research Project

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Land-Cover and
Land-Use Change Program

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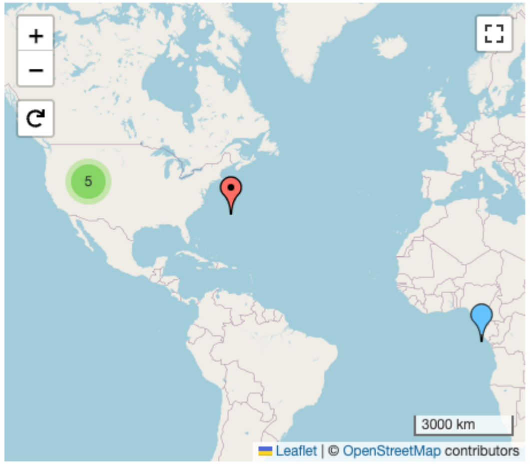
A Remote Sensing Analysis of Heat Stress, LCLUC, and Women's Health in Sub-Saharan Africa

Start Date	01/01/2022
End Date	03/19/2026
Grant Number	80NSSC24K0531
Science Theme Name	Detection and Monitoring of LCLUC Climate Variability and Change Vulnerability, Adaptation Observations and Data Drivers of Change
Region	Africa West Africa
Solicitation	NNH21ZDA001N - LCLUC



NASA

Land-Cover and Land-Use Change Program

BOSTON UNIVERSITY

INTERNATIONAL FOOD POLICY RESEARCH INSTITUTE
IFPRI

MONTANA STATE UNIVERSITY

The Research Team

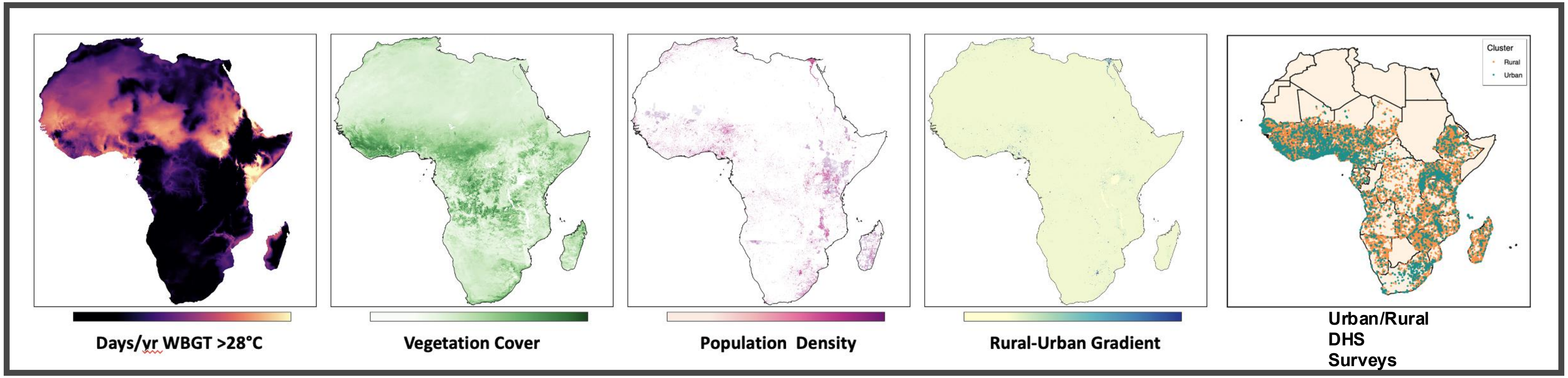
Nina Brooks	Principal Investigator	Boston University, Boston, USA **(Now University of Michigan)**
Cascade Tuholske	Co-Investigator	Montana State University, Bozeman, USA
Kwaw Andam Oliver Kirui Hyacinth Edeh	Collaborator	International Food Policy Research Institute, Abuja, Nigeria
Chris Funk	Collaborator	University of California, Santa Barbara, USA
Kathryn Grace	Collaborator	University of Minnesota, Minneapolis, USA
Shraddhanand Shukla	Collaborator	University of California, Santa Barbara , USA

Land-Cover and Land-Use Change Program

Knowledge Gaps

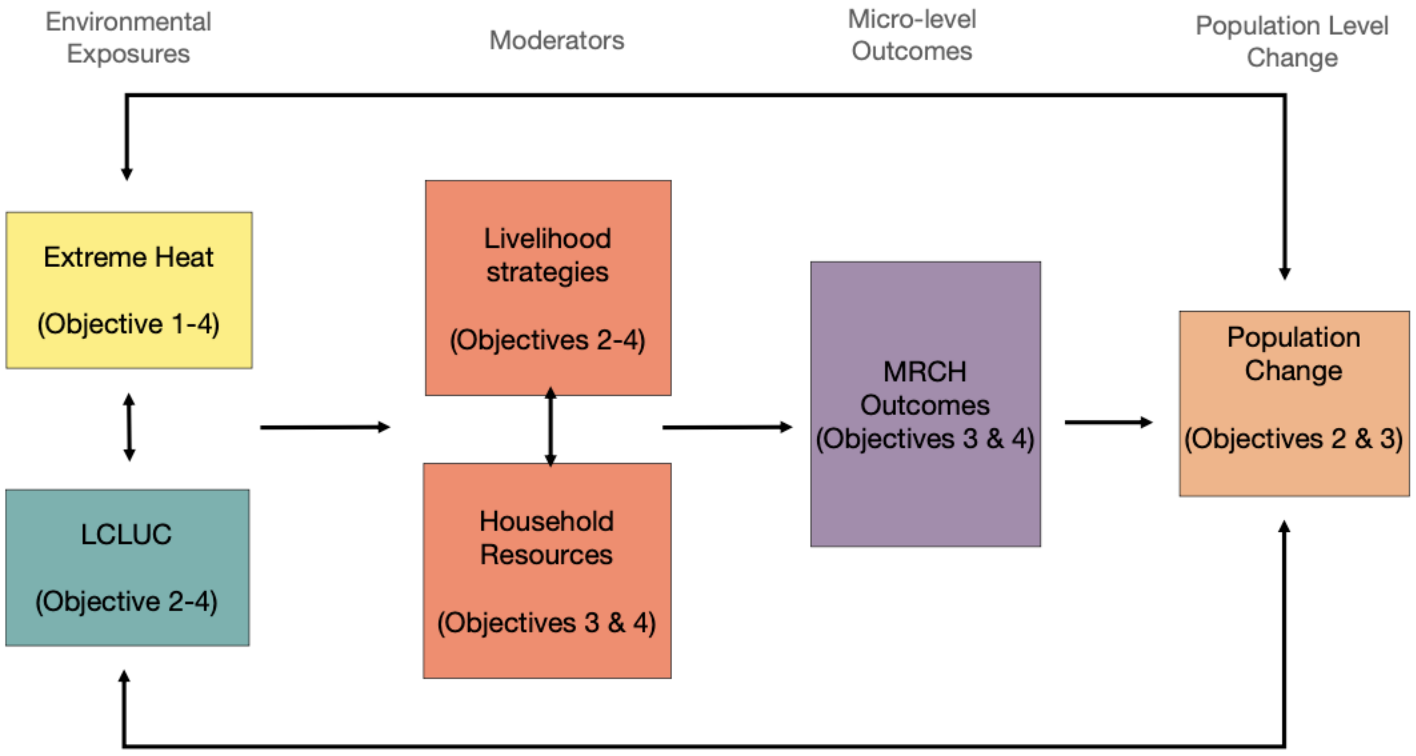
- Climate change is driving an increase in dangerous heat extremes across Sub-Saharan Africa.
- Land-cover and land-use change (LCLUC) can amplify or mitigate extreme heat, depending on the local context.
- Extreme heat impacts maternal and child heat, but little known about such impacts outside of North American and European contexts.
- What are the relationships between climate-change-drive extreme heat, LCLUC, and **maternal and child health outcomes (MCH)** across Sub-Saharan Africa?
- What policies are effective for reducing harm from extreme heat?

Research Objectives



1. Measure and map changes in extreme heat event characteristics and exposure
2. Identify interactions hotspots between extreme heat events characteristics exposure, and LCLUC
3. Quantify the impacts of heat-stress and LCLUC on maternal reproductive and child health
4. Conduct a High-Resolution Case Study for Nigeria

Research Objectives



Adapted from Grace 2017

Figure 1: Conceptual Diagram of Research Components

Obj 1 - Measure and map changes in extreme heat event characteristics and exposure

Data advancements

- CHIRTS-daily most accurate, high-resolution (0.05°) daily temperature record from which we can measure changes in duration, frequency, and intensity of both hot-humid and hot-dry extreme heat events from 1983 - 2016.
- Long term Global Land Change (0.05°) measures annual fractional land cover for tree canopy, short vegetation, and bare ground from 1982 - 2016 using the AVHRR satellite record.
- Global Human Settlement Layer SMOD and Pop datasets provide settlement type and populations for 1975, 1990, 2000, and 2015.

Research advancements

- Map how hot-humid and hot-dry heat waves varied by land cover and demographic change across Africa.

Obj 2 - Map interactions hotspots between extreme heat exposure and LCLUC

Methodological advancements

- Apply & compare standard approaches and **machine learning** approaches to identifying LCLUC-heat hotspots.
- Explore different **spatial & temporal scales for identifying hotspots**, given the long time series of the data.

Research advancements

- Link hotspots to maternal reproductive and child health outcomes to see how “hotspot” areas fare compared to “average” areas.

Obj 3 - Assess the impacts of heat-stress and LCLUC on maternal reproductive and child health

Research advancements

- Quantify humid-heat and LCLUC **impacts on child health & pregnancy outcomes** (e.g., infant mortality, stunting/wasting, and birthweight).
- Understand how (stated) fertility intentions and fertility control (via contraceptive use) are affected.
- Examine **cumulative shocks/repeated exposures** and maternal reproductive and child health outcomes.
- Identify **policy strategies** for improving maternal and child health in the context of climate change.

Obj 4 - Conduct a High-Resolution Case Study for Nigeria

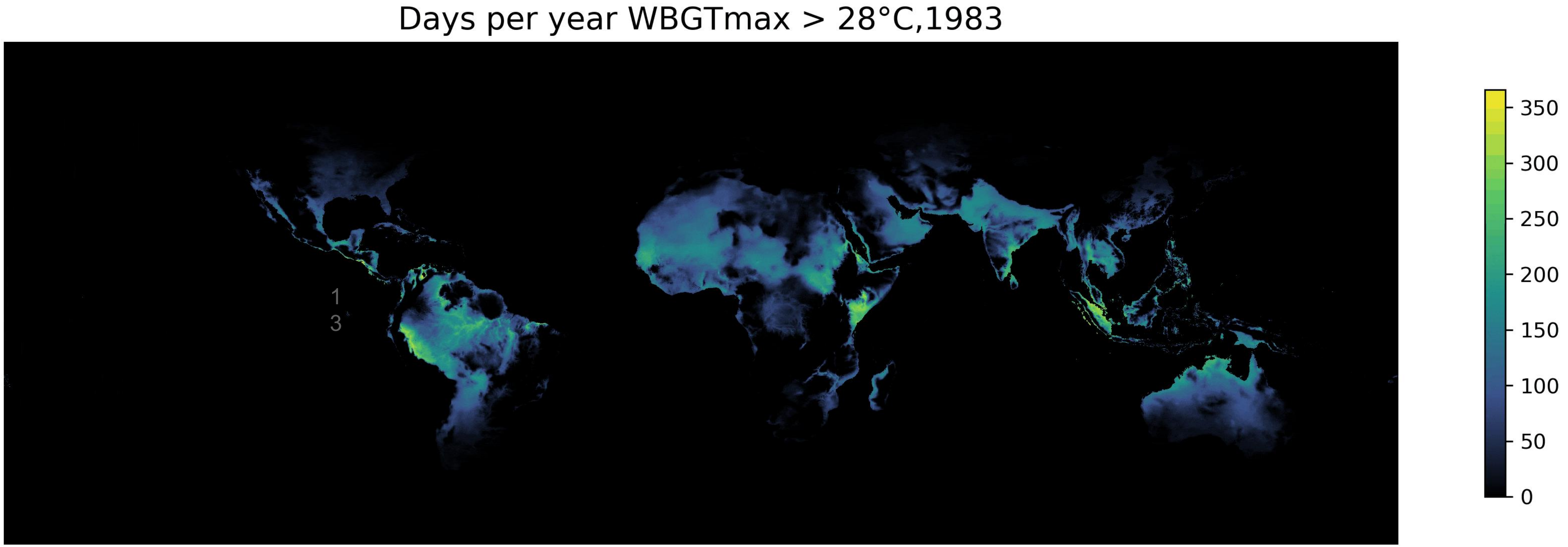
Data Advancements

- From a remote-sensing standpoint, the Africa-wide LCLUC-heat-population analysis is coarse-grained.
- Leverage higher-resolution LCLUC processes (Landsat, Worldview, PlanetScope etc.) to construct a high-res dataset of heat and LCLUC.

Research advancements

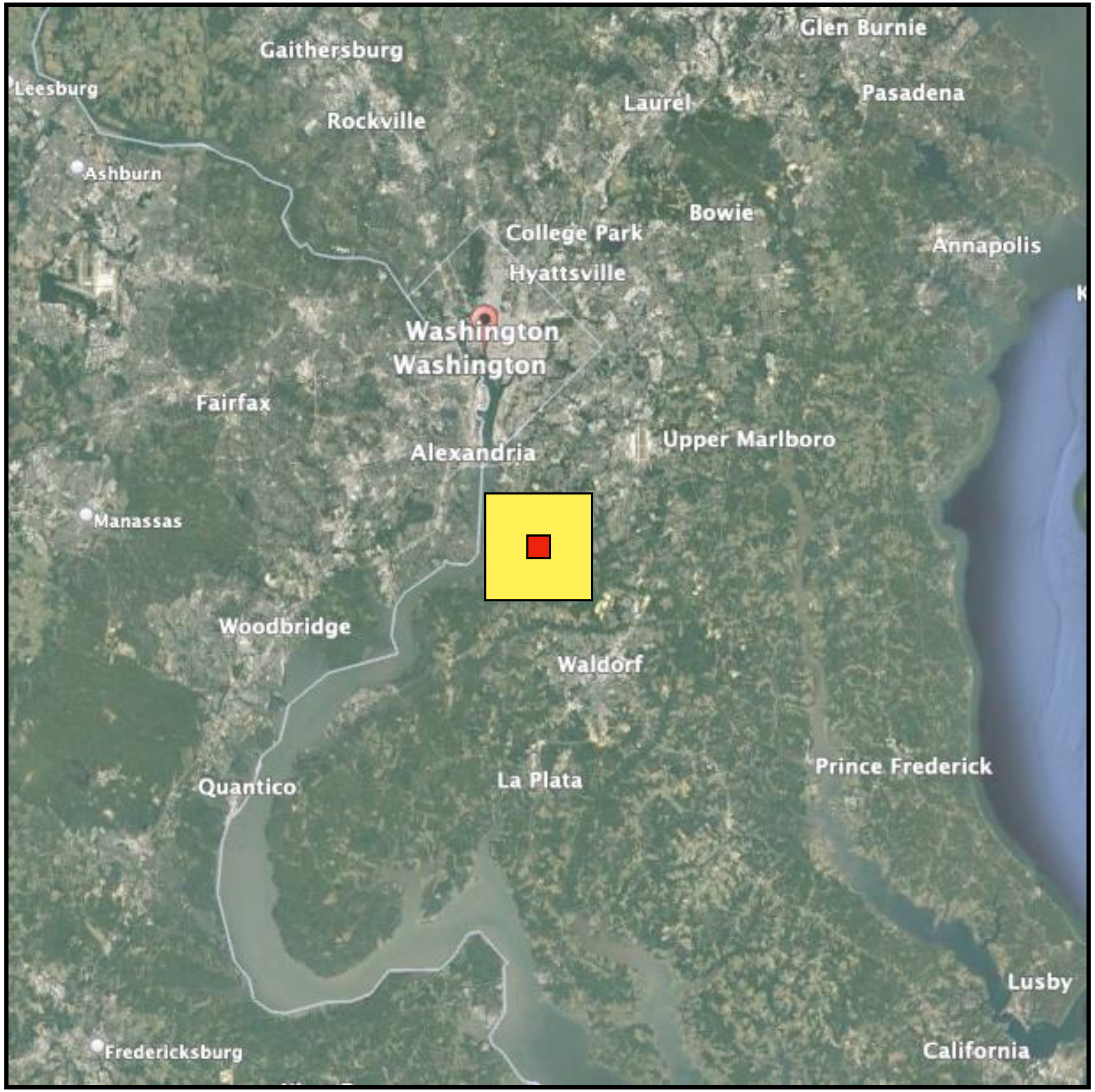
- A 5 x 5 km pixel may have the same LCLUC-heat-population dynamics. But the **sub-pixel LCLUC dynamics may be heterogeneous**.
- For example, examine how increased irrigation practices correlate with long-term increases in hot-humid heat & how this relates to maternal/child health outcomes in Nigeria.

Datasets: Temperature



- CHIRTS-daily: High-resolution (°0.05) observational estimates of daily maximum air temperatures and daily maximum wet-bulb globe temperature (WBGTmax)
- Freely available from UC Santa Barbara Climate Hazards Center

Datasets: CHC-CMIP6 projections



Williams *et al.* 2024

Current CMIP-6
Climate Projection
100 x 100 km

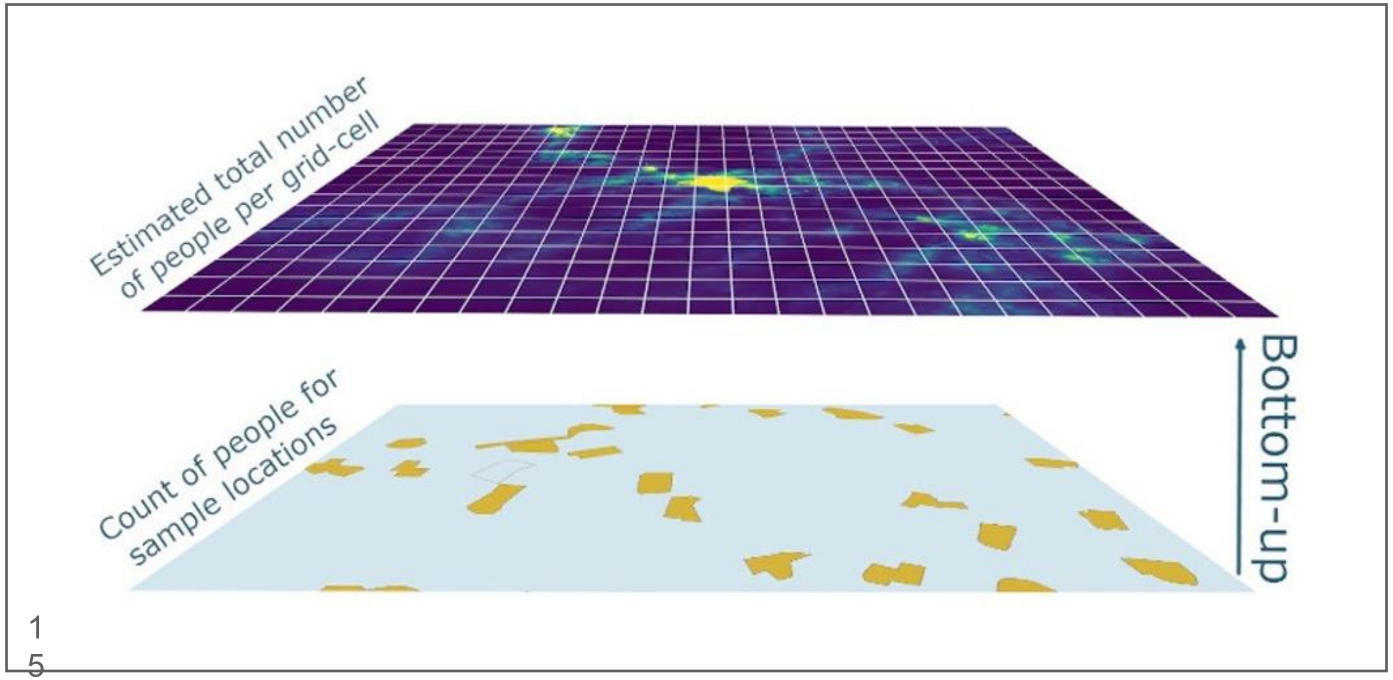
ERA-5
Temperature
25 km X 25 km

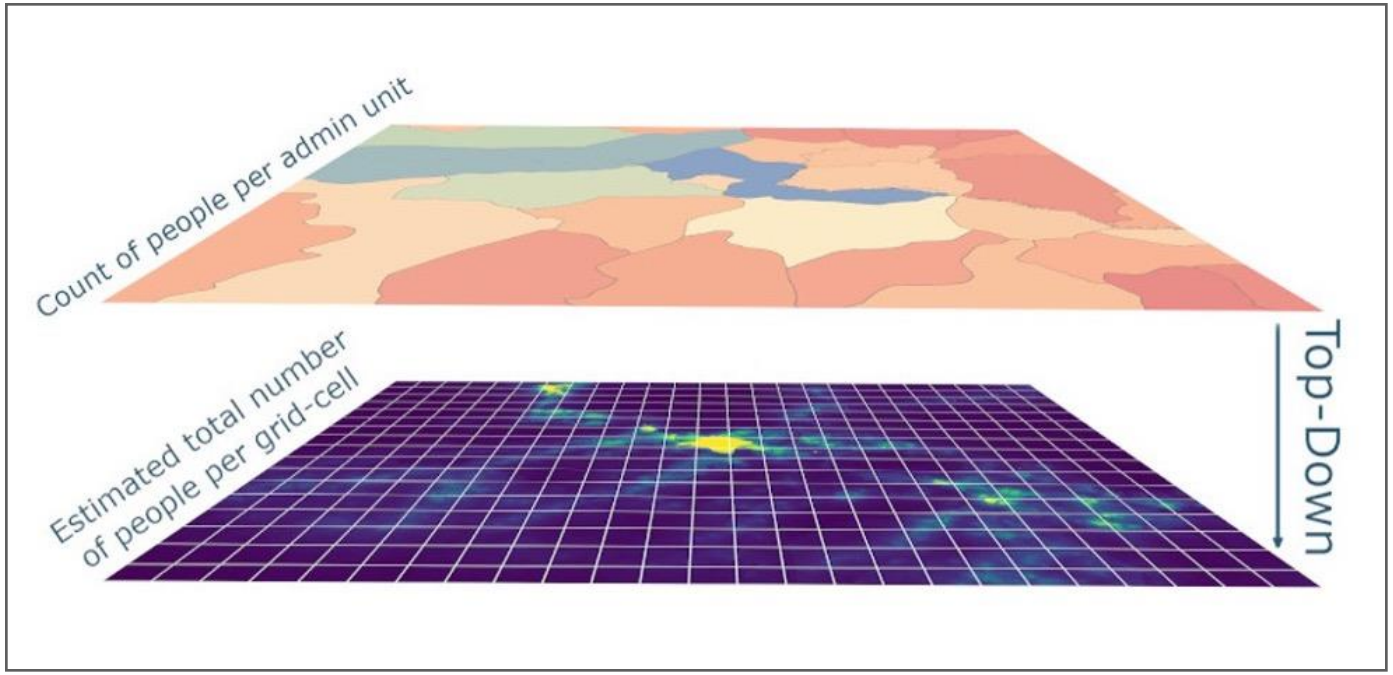
New **CHC-CMIP6**
2030 & 2050
Daily Climate
Projections
5 x 5 km

Led by
Emily Williams,
UC Santa Barbara
(Now UC Merced)

Free via
UCSB CHC

Datasets: Gridded Population Data





- Blend satellite data with best-available census to put people into boxes.

- Latest gridded population data is available at 100m or less.

- Still issues with accuracy
- (e.g. undercounting of urban poor).

- Available from PopGrid

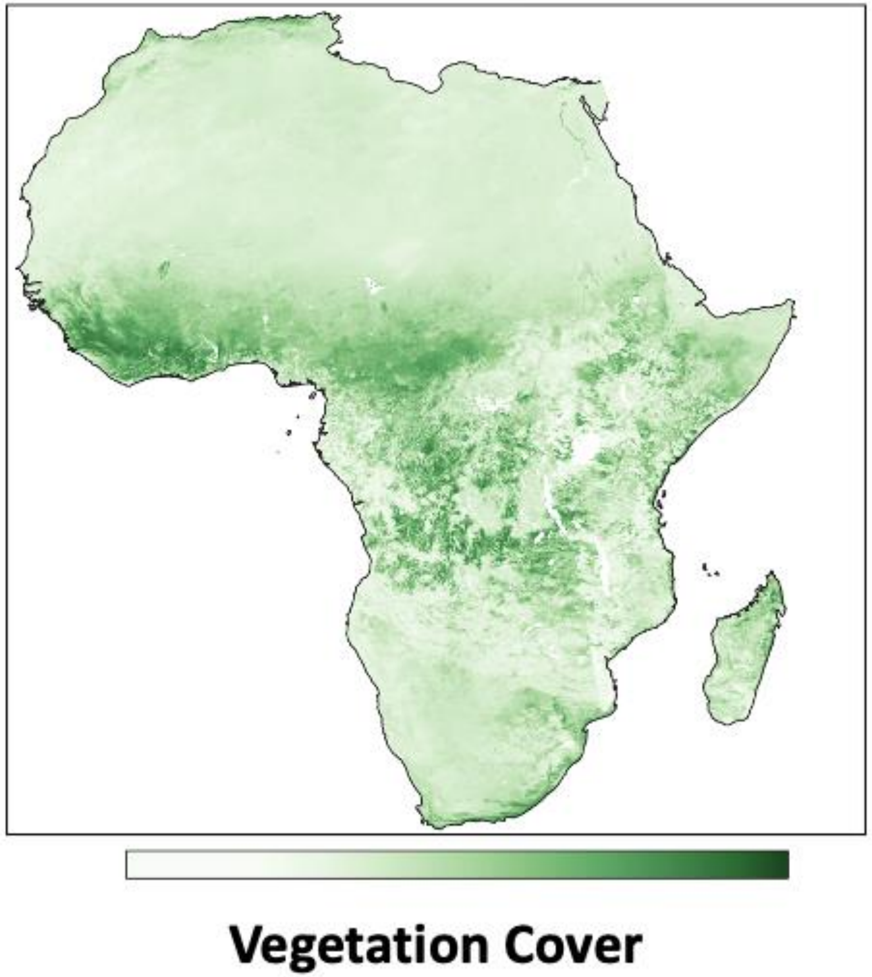
Datasets: Land-Cover and Land-Use Change

GLAD Long-term Global Land Change dataset

- 1983- 2016, 5-km
- Percent short vegetation, tree canopy, and bare ground
- Aligns with CHIRTS-daily temperature record

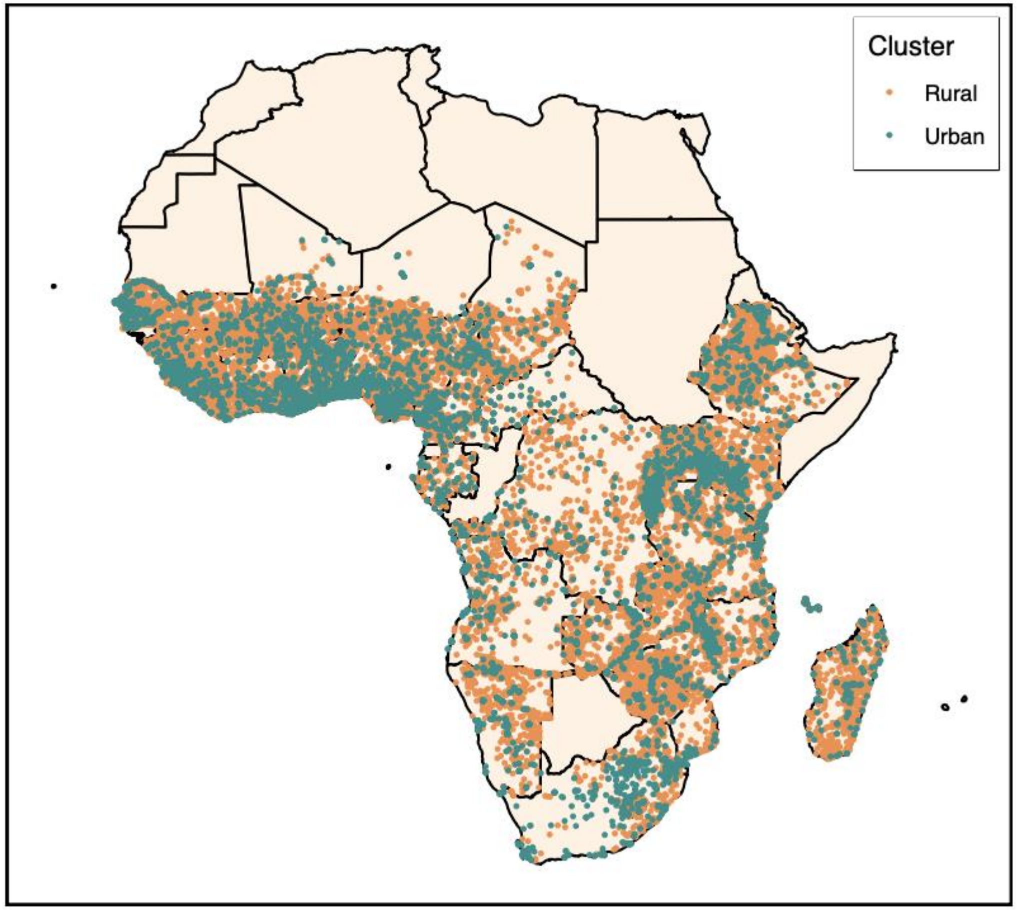
Future Datasets

- Landsat / Sentinel
- PlanetScope (3m)



Datasets: Demographic Health Surveys (DHS)

- Standardized demographic surveys conducted in over 90 low-and-middle income countries
- Many contain geographic information
- Freely available to download from IPUMS DHS
- Download from web browsers or with R package
- Nigeria: 6 waves available (1990, 1999, 2008, 2013, 2018)
- Extensive data on population, health, HIV, and nutrition, fertility and family planning, malaria, and more



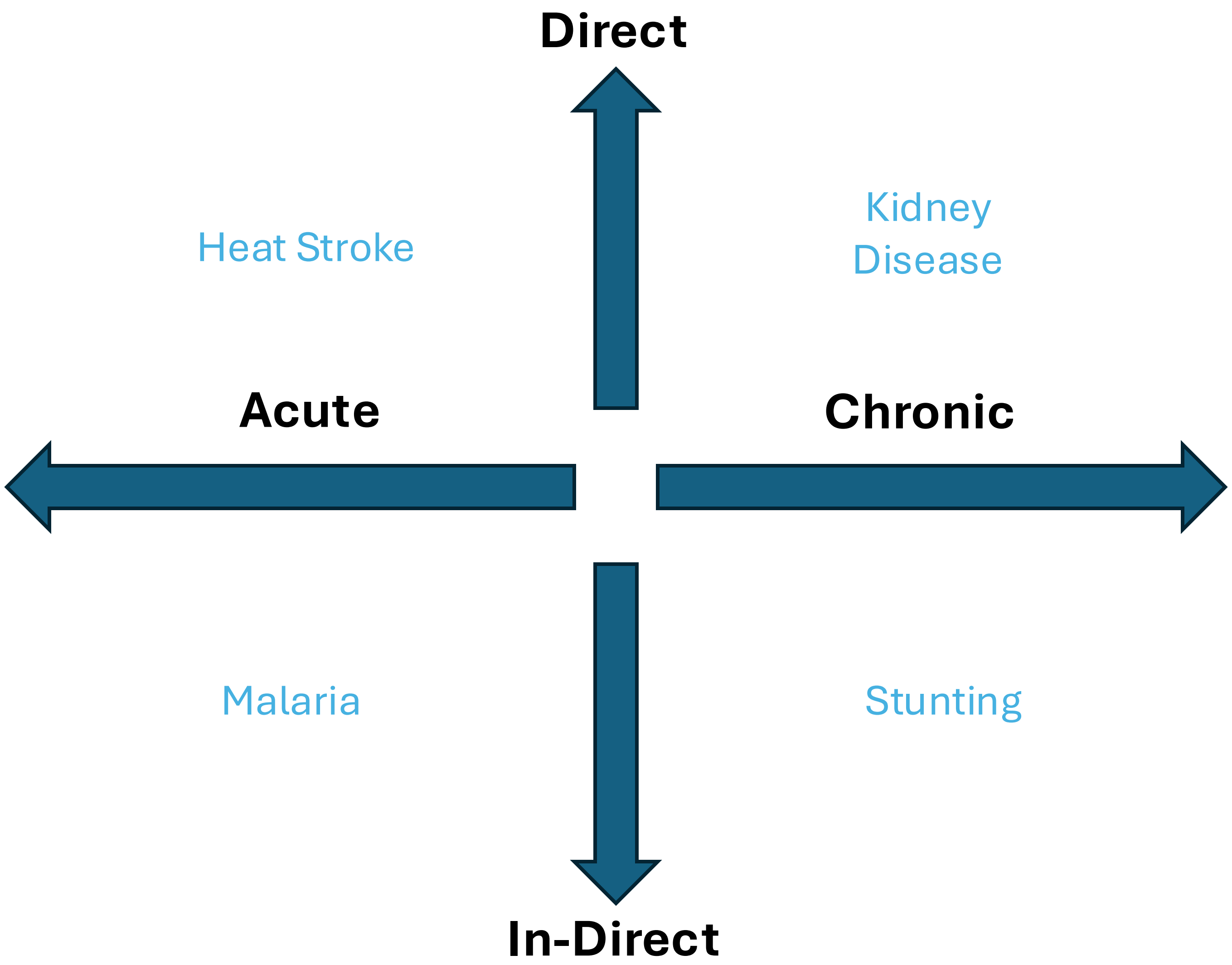
Heat Impacts to Human Health

Brainstorm: how does heat impact human health?

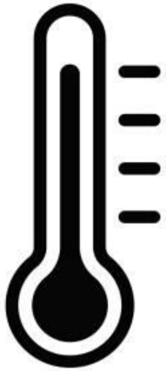
Things to Consider

- Where does exposure occur?
- When does exposure occur?
- What are the mechanisms that cause harm to health and well-being?
- What socioeconomic, cultural, economic, demographic, and political factors mediate or amplify impacts?

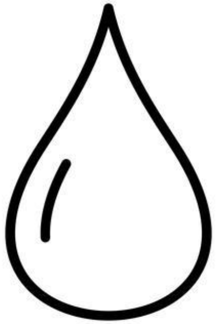
How does heat impact human health?



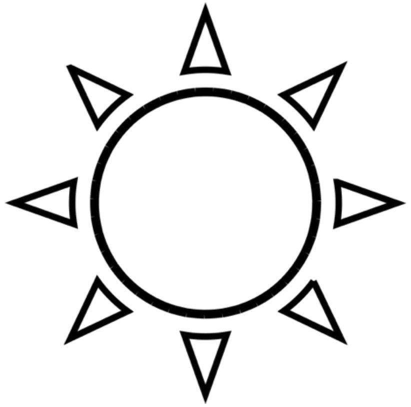
Meteorological factors that influence heat stress.



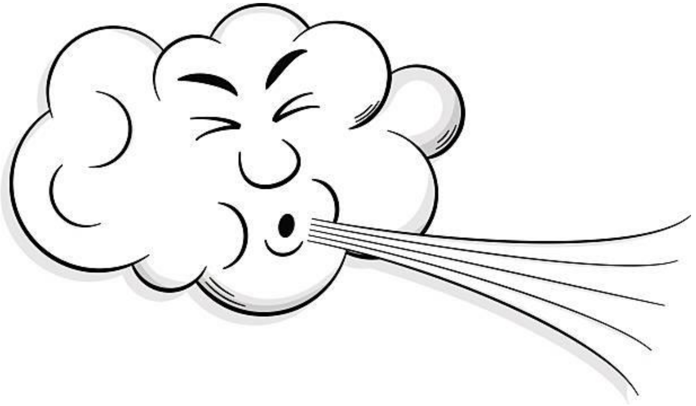
(Dry)
Air Temperature



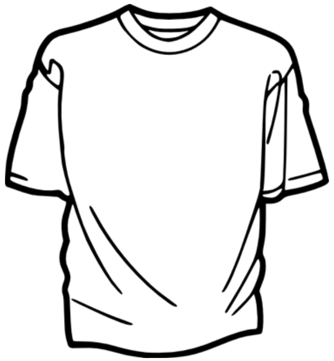
Atmospheric
Moisture



Radiation



Wind

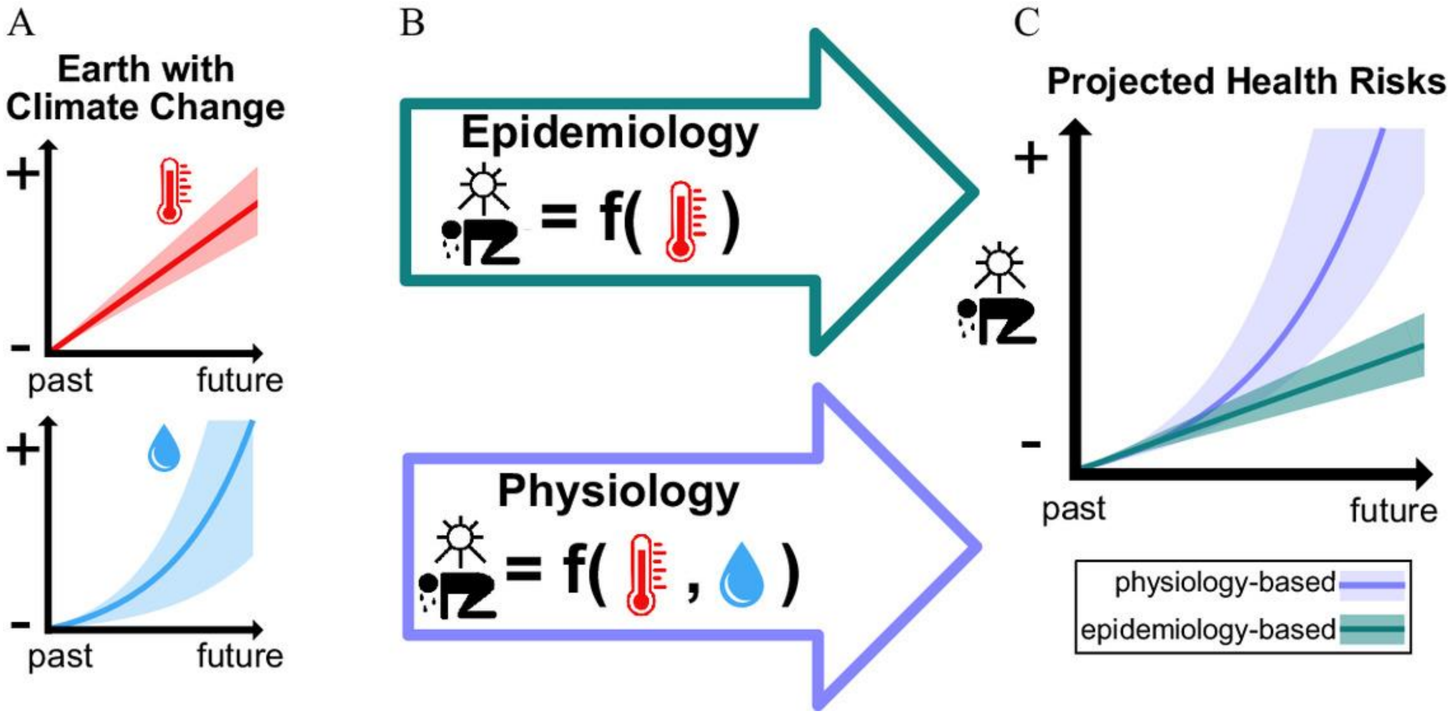


Clothing

Heat stress metrics

- There are over 300 different metrics in use to define heat stress.
- There is no “official” definition of a heat wave.
- Important to be clear why a heat stress metric is being selected and how it is constructed.
- Recommend using several metrics in environmental epidemiological analysis when possible.
- Today we will cover six metrics: air temperature, heat index, humidex, wet-bulb temperature, wet-bulb globe temperature, and universal thermal comfort index

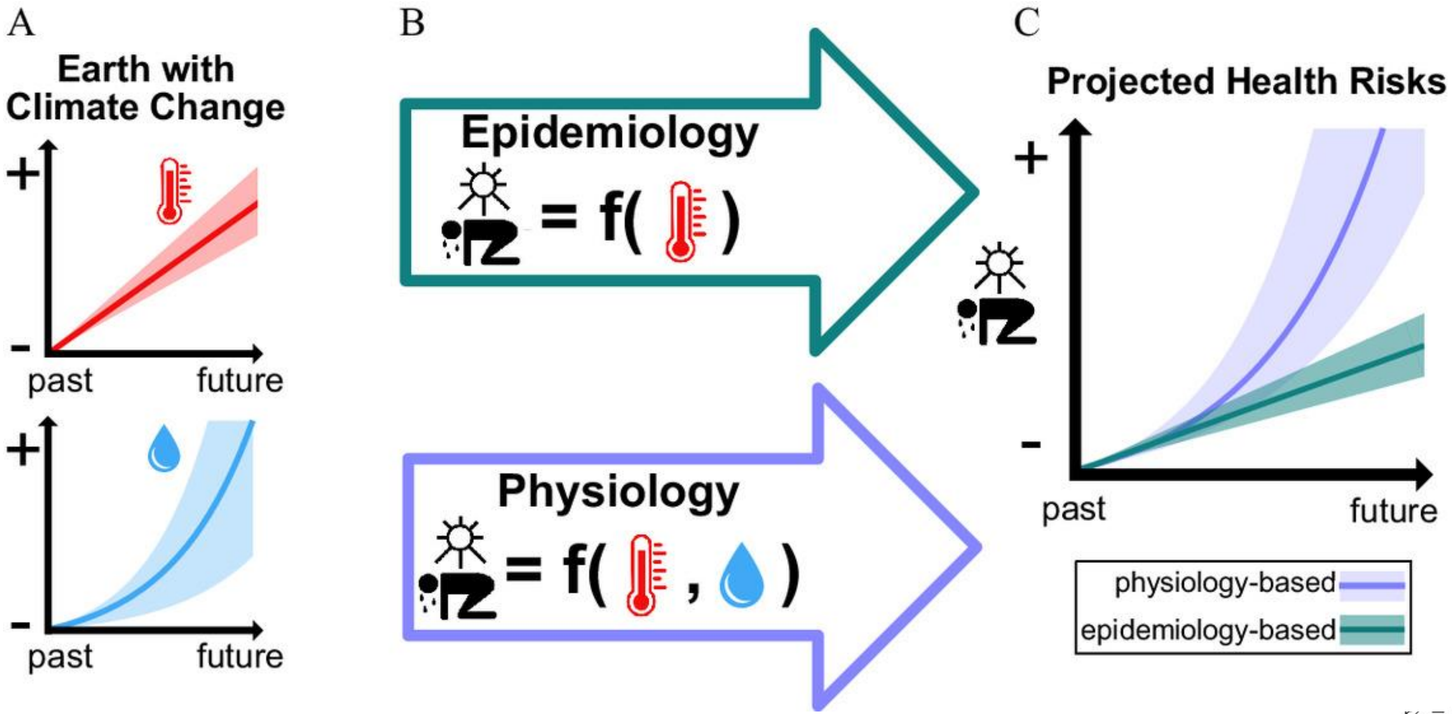
Heat & Health



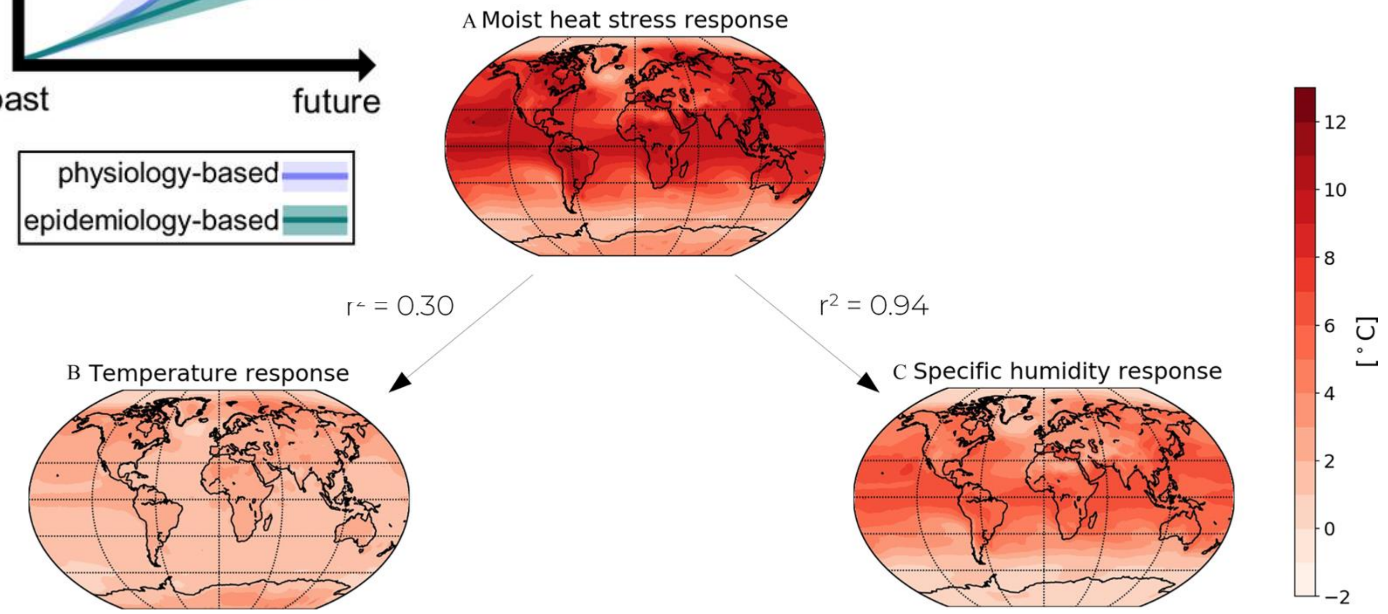
- **Actual air temperature** measured by a standard thermometer freely exposed to the air **<u>but shielded from moisture and radiation.</u>**
- Same as "dry-blub temperature", "surface air temperature" or "2-meter air temperature (T2m)"
- Most widely meteorological observation available
- Psychrometric – a property of thermal dynamics

Baldwin *et al.* 2023

Heat & Health



The relationship between moisture and temperature is really important for climate projections due to **<u>exponential scaling of humid-heat.</u>**



Baldwin *et al.* 2023

Importance of Spatial Scale

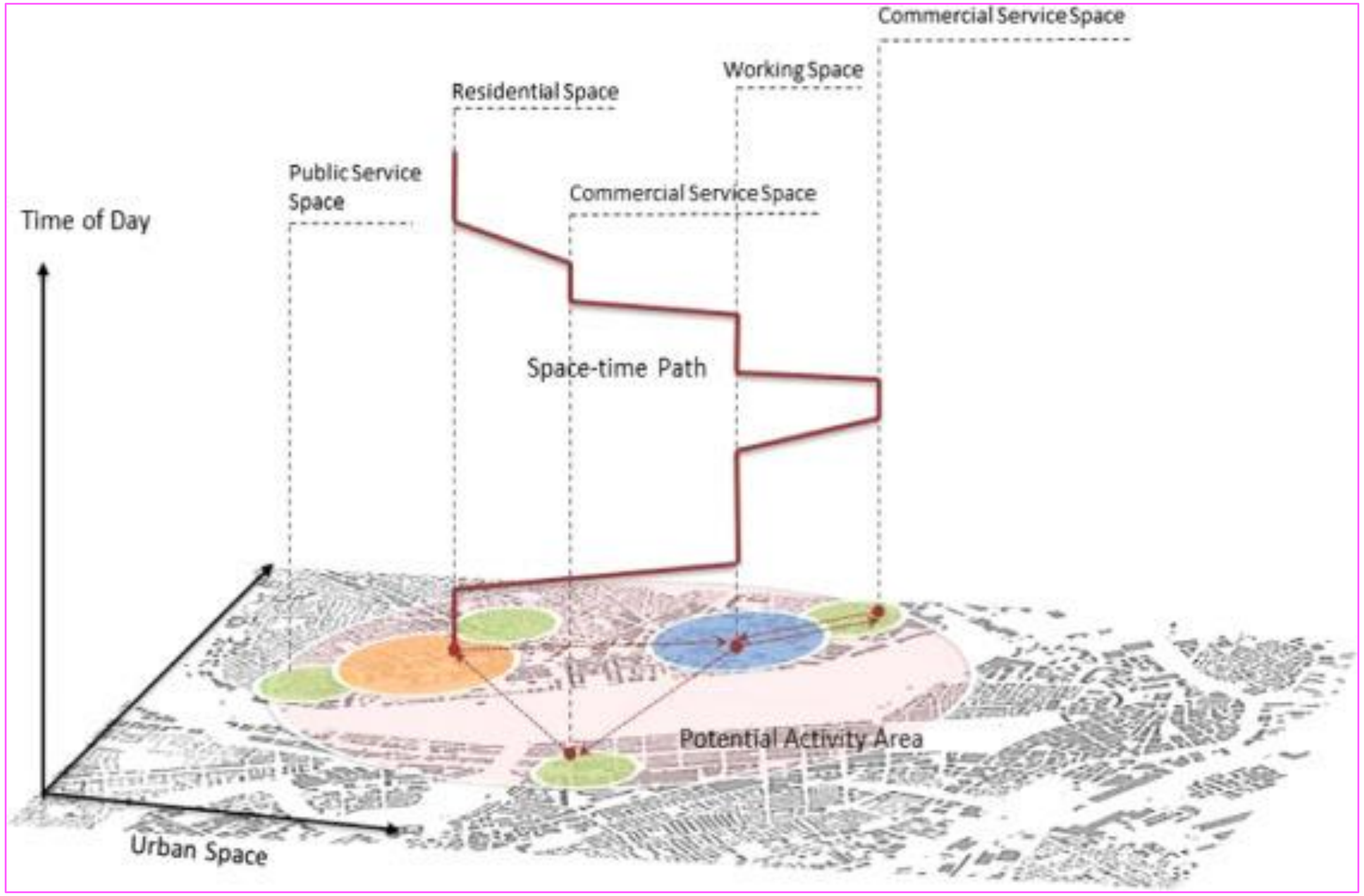
<u>Do we have the human location data to measure ...</u>

- How long was a person exposed?
- Where was a person exposed?

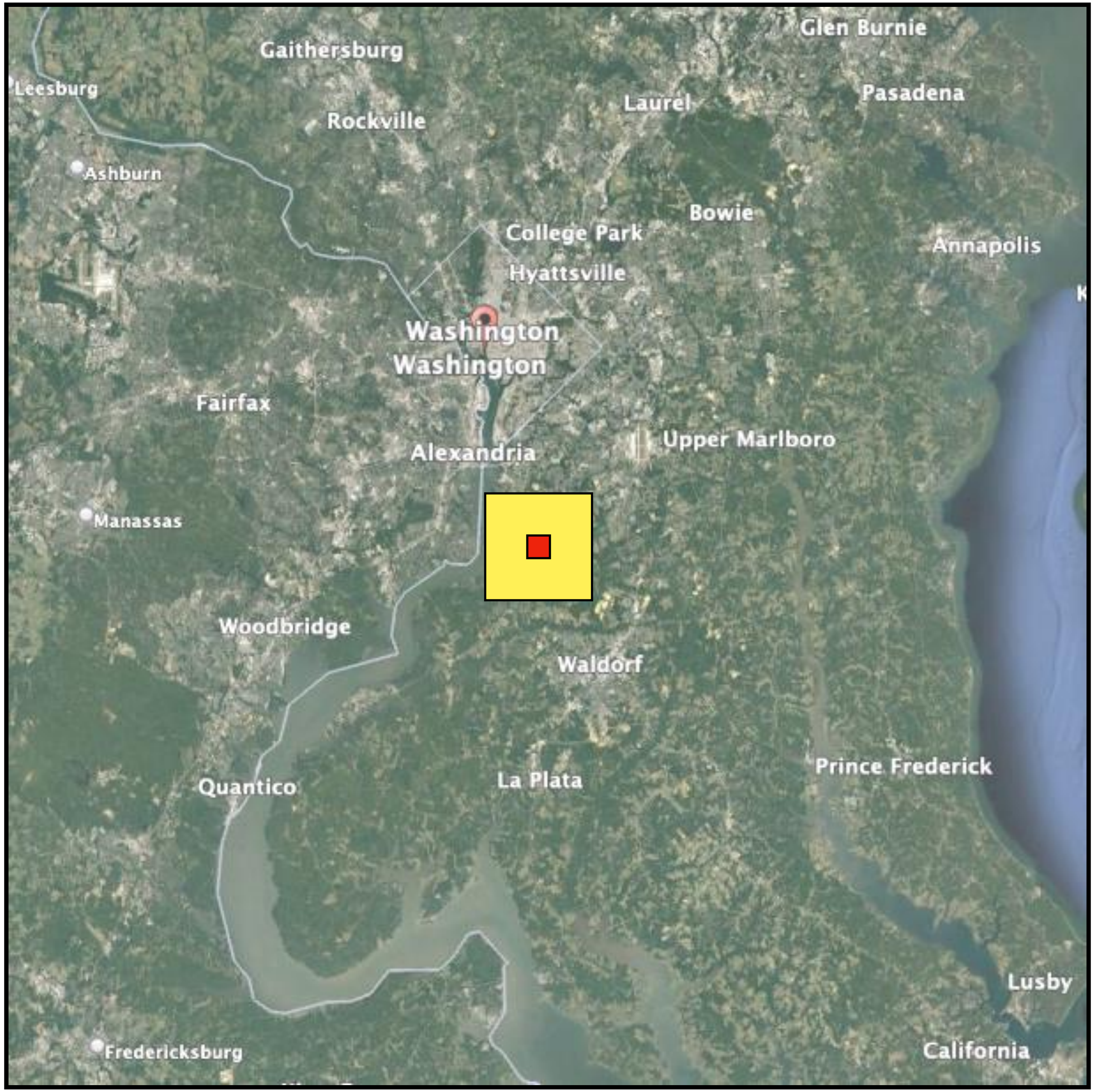
<u>Do we have the climate location data to measure ...</u>

- How long was a person exposed?
- Where was a person exposed?

Space-Time Prism



Importance of Spatial Scale



Williams *et al.* 2024

Current CMIP-6 Climate Projection 100 x 100 km

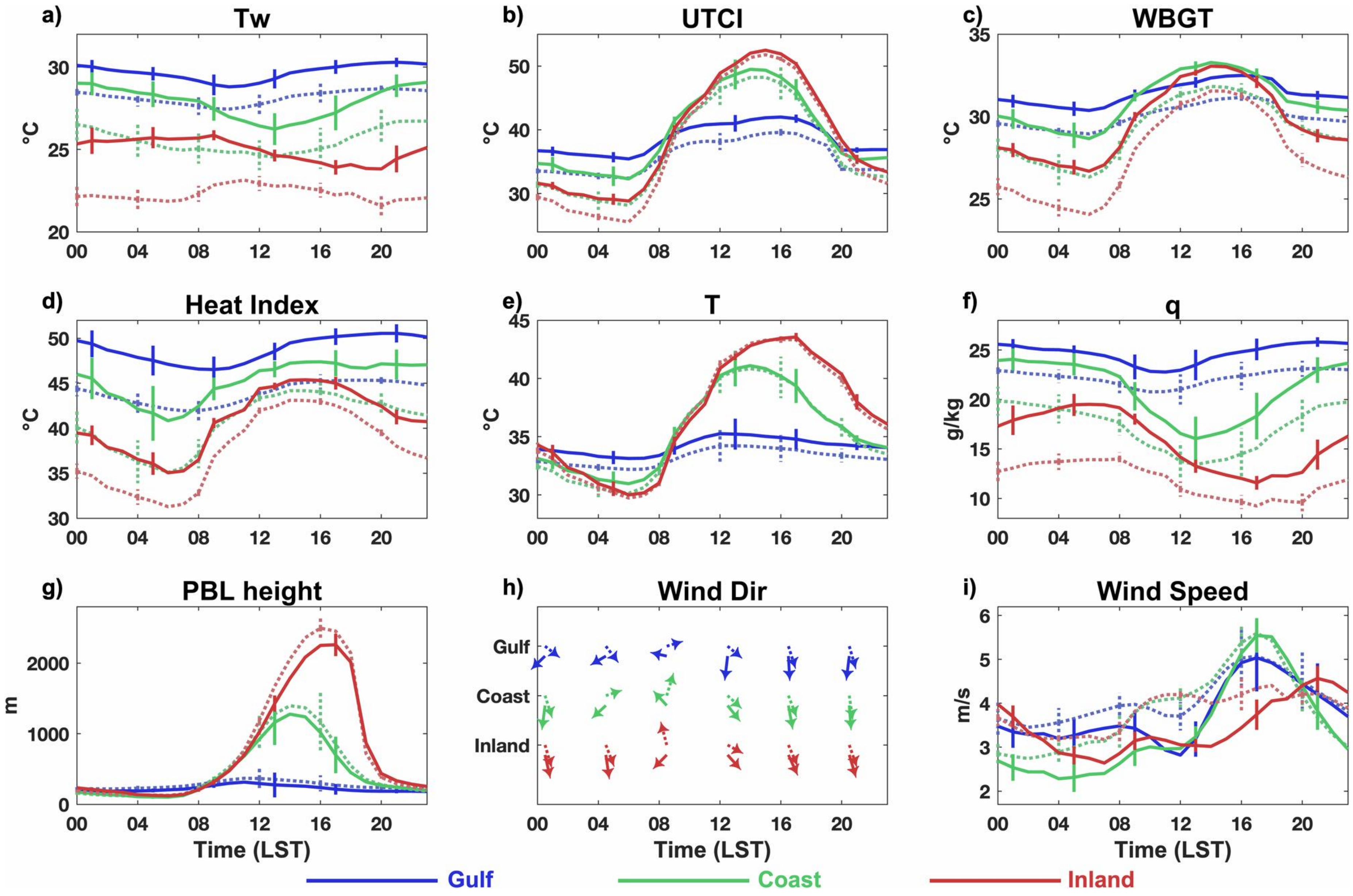
ERA-5 Temperature 25 km X 25 km

New **CHC-CMIP6** 2030 & 2050 Daily Climate Projections 5 x 5 km

Led by **Emily Williams,** UC Santa Barbara (Now UC Merced)

Free via UCSB CHC

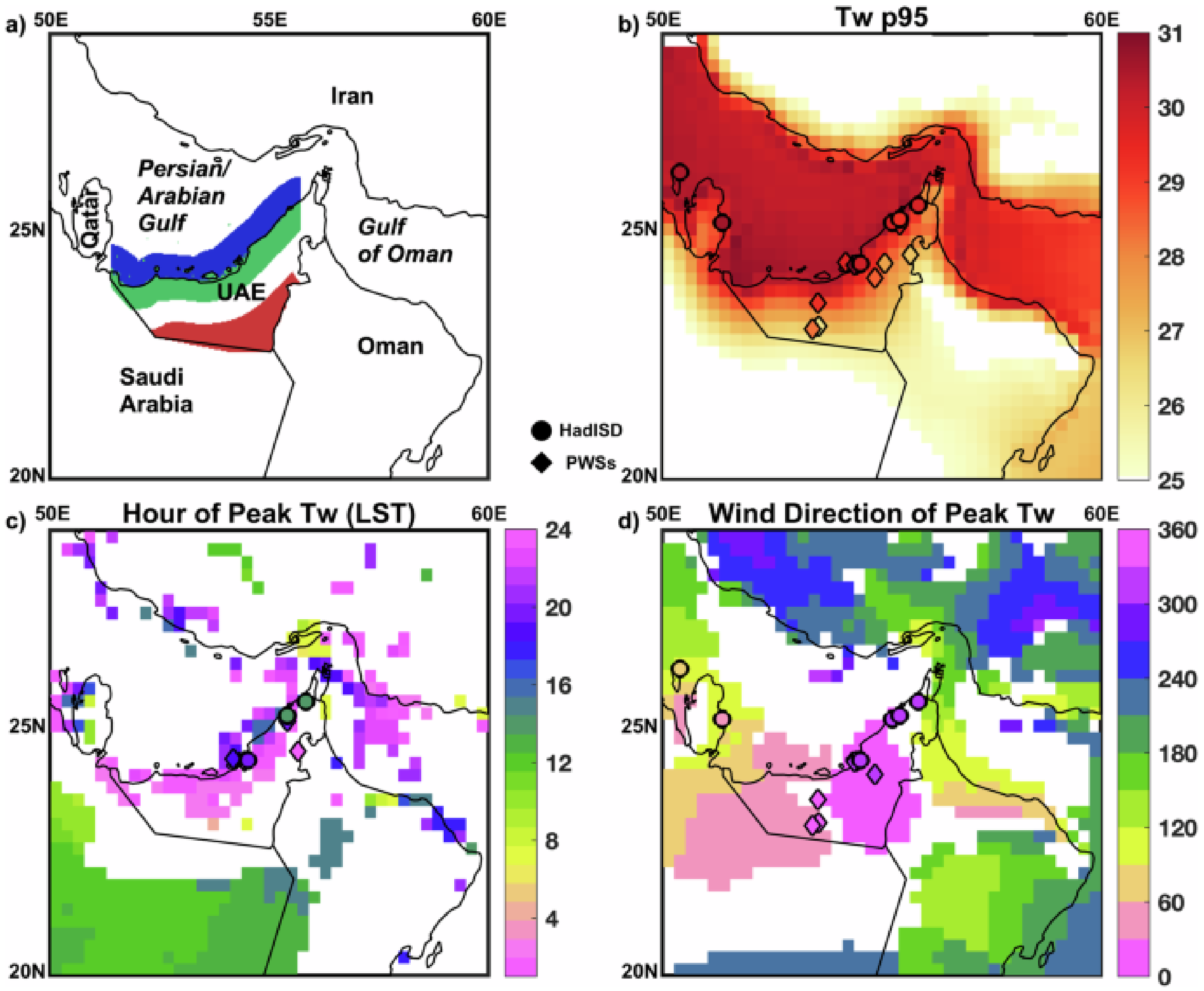
Importance of Temporal Scale



Raymond *et al.* 2024

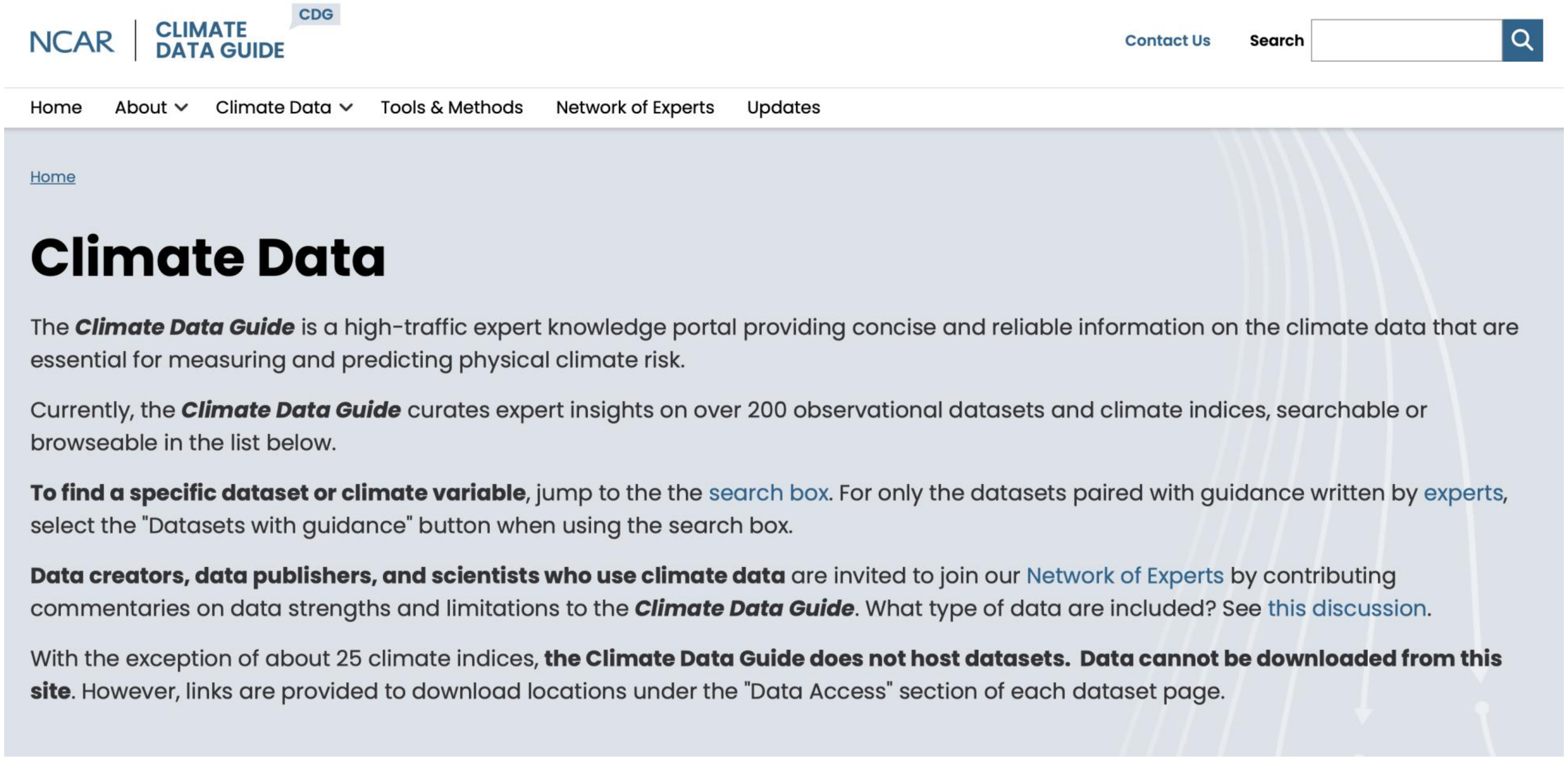
The Diurnal Cycle

- Must align meteorological inputs in the diurnal cycle to construct heat stress metrics like WBT or UTCI.
- If you have hourly data … estimate metric at each hour, then average all 24 hours or extract the max/min for a given day.
- Or using daily minimum moisture and maximum temperature
- **Always be clear how daily minimum, maximum and/or average are calculated**



Raymond *et al.* 2024

Observational Climate Data



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Climate Data

The ***Climate Data Guide*** is a high-traffic expert knowledge portal providing concise and reliable information on the climate data that are essential for measuring and predicting physical climate risk.

Currently, the ***Climate Data Guide*** curates expert insights on over 200 observational datasets and climate indices, searchable or browseable in the list below.

To find a specific dataset or climate variable, jump to the the search box. For only the datasets paired with guidance written by experts, select the "Datasets with guidance" button when using the search box.

Data creators, data publishers, and scientists who use climate data are invited to join our Network of Experts by contributing commentaries on data strengths and limitations to the ***Climate Data Guide***. What type of data are included? See this discussion.

With the exception of about 25 climate indices, **the Climate Data Guide does not host datasets. Data cannot be downloaded from this site**. However, links are provided to download locations under the "Data Access" section of each dataset page.

https://climatedataguide.ucar.edu/climate-data

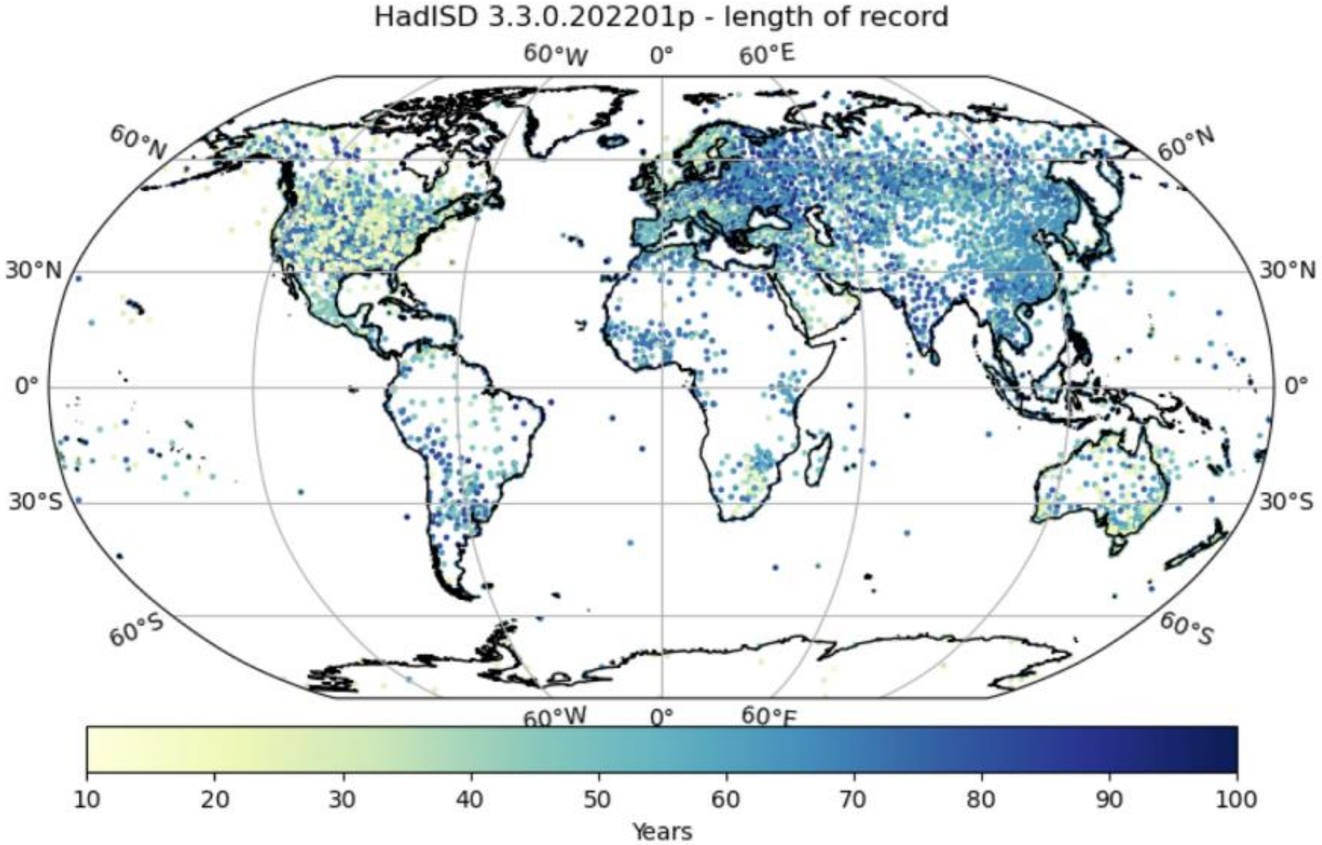
Weather Station Data

- Local stations (try asking your National MET service)
- Berkeley Earth
  - https://berkeleyearth.org/data/
  - About 50,000 Individual station data used to produce monthly record.
- HadISD Stations:
  - https://www.metoffice.gov.uk/hadobs/hadisd/index.html
  - Quality controlled sub-daily data from NOAA NCEI's Integrated Surface Database.
  - About 10,000 stations.
  - 1931 – present
  - Few stations for Africa
- CRUTEM5 Temperature station data:
  - https://www.metoffice.gov.uk/hadobs/crutem5/data/CRUTEM.5.0.2.0/download.html
  - monthly-mean temperature records from 10639

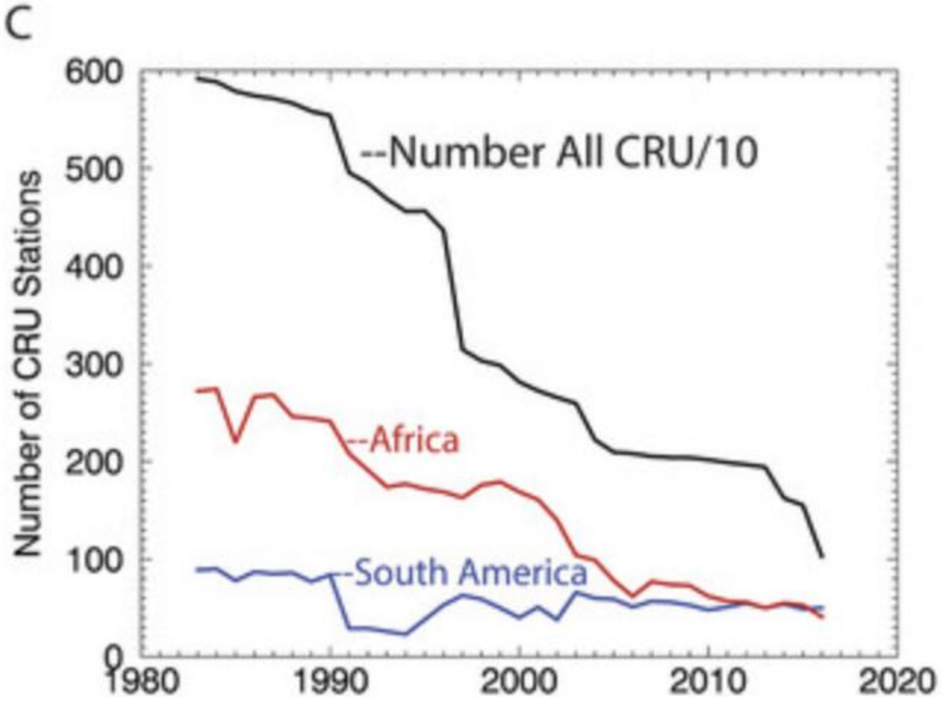
Weather Station Data

- Weather station data is not usually available in a use-friendly format.

- Quality control is important.
  - Station go offline
  - Stations don't all have the same reporting time-period
  - Stations get moved.



Weather Station Data



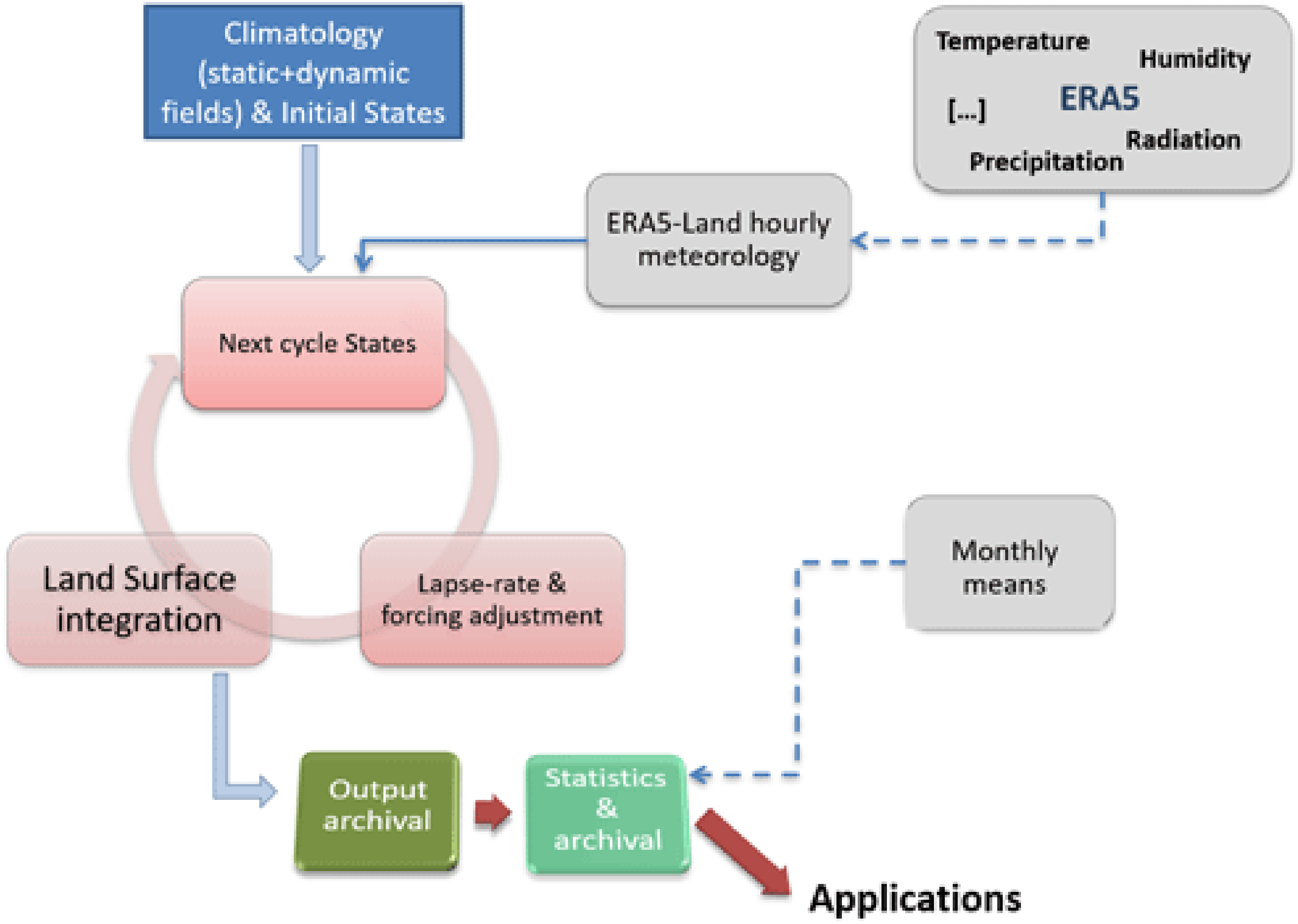
Gridded Observation Datasets

- NOAA Climate Prediction Center Global Unified Temperature
  - https://psl.noaa.gov/data/gridded/data.cpc.globaltemp.html
  - 0.5° spatial resolution (very course-grained)
  - 1979 – present
  - Daily min & max 2m air temperature

- CHIRTS-daily
  - https://www.chc.ucsb.edu/data/chirtsdaily
  - Blends satellite data and station data to down-scale ERA5 (more later)
  - 0.05° (about 5km) spatial resolution
  - Daily min & max 2m air temperature
  - 1983 – 2016 (experimental version to present available)

Climate Reanalysis Data

- Weather models use “initial conditions” (e.g. observations) to forecast the near-term weather.
- Reanalysis products use “initial conditions” to hindcast the weather.
- Produce sub-daily (hourly) ocean, land, and atmospheric conditions.



Climate Reanalysis Data

Four main global reanalysis products:

- ERA5 & ERA5-Land produced by the European Union
  - https://www.ecmwf.int/en/research/climate-reanalysis
  - Since 1950, Hourly at 25 and 9 km respectively
- MERRA2 produced by NASA:
  - https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/
  - Since 1980 6 hours, 0.5° latitude × 0.625°
- NCEP/NCAR Reanalysis:
  - https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html
  - Since 1948, 6 hours, 1.875 × 1.904129 gaussian grid
- JRA-55:
  - https://data.ucar.edu/dataset/jra-55-japanese-55-year-reanalysis-daily-3-hourly-and-6-hourly-data
  - 1958 onward, 3-hour, 1.55°

Climate Reanalysis Data

Four main global reanalysis products:

- ERA5 & ERA5-Land produced by the European Union
  - https://www.ecmwf.int/en/research/climate-reanalysis
  - Since 1950, Hourly at 25 and 9 km respectively
- MERRA2 produced by NASA:
  - https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/
  - Since 1980 6 hours, 0.5° latitude × 0.625°
- NCEP/NCAR Reanalysis:
  - https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html
  - Since 1948, 6 hours, 1.875 × 1.904129 gaussian grid
- JRA-55:
  - https://data.ucar.edu/dataset/jra-55-japanese-55-year-reanalysis-daily-3-hourly-and-6-hourly-data
  - 1958 onward, 3-hour, 1.55°

A few points on historical climate data

- All historical observational and reanalysis data will have error, reporting gaps, and uncertainty.

- It’s important to be aware of this uncertainty.

- Some teams are preforming their analysis using multiple dataset to triangulate the results.

- When in doubt, contact a data producer or reach out to someone who is an expert in that specific dataset.

Coughlan de Perez *et al.* 2023

A few points on historical climate data

Stations

- Advantages: spatial precision, temporal record “ground-truth”, unmodeled
- Disadvantages: low spatial coverage, data gaps, stations move, recording errors

Gridded Products

- Advantages: consistent model; global coverage,
- Disadvantages: biased, course-resolution, assimilation errors

File Types

GeoTIFF (.tif)

- Usually 2 dimensional with (lat x long)
- Each file is 1 time slice and one meteorological variable (e.g. air temperature)
- Commonly used with GIS software like QGIS
- Coordinate reference system and projection important

NetCDF (.nc)

- 3-dimentional data cube (time x lat x long)
- Each file can have more than one meteorological variable (e.g. air temperature and precipitation)
- Coordinate reference system and projection important

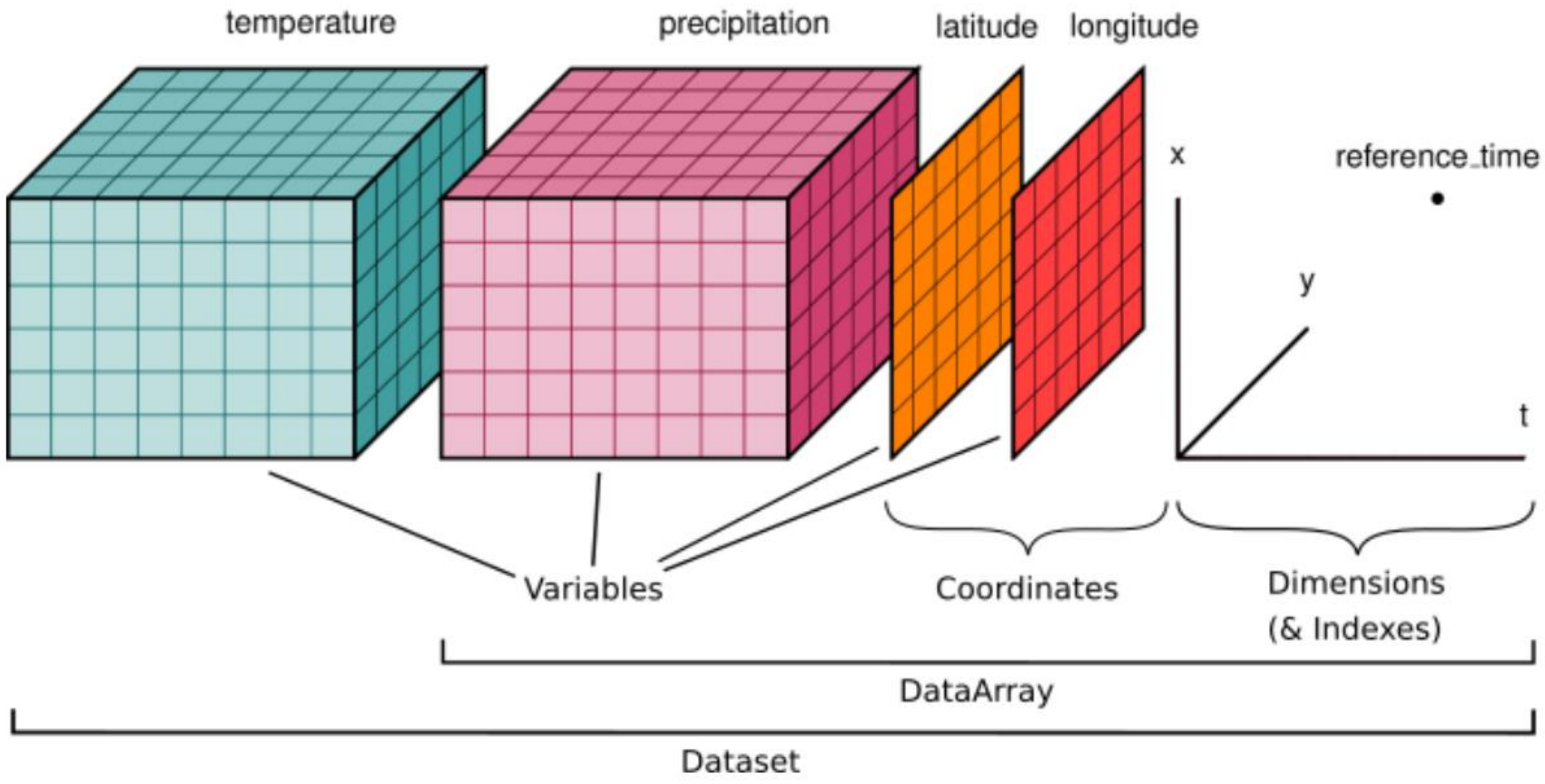
Data Extraction Workflow

- Xarray Python package to creates stacked, 3d data arrays that uses labels for all data and coordinates.

- Very easy to subset and extract climate by time and by location.

- All we need are the latitude and longitude of the clusters

- Extract data with just a few lines of code.



Data Extraction Workflow

- ERA5 Hourly Climate Reanalysis

- Download hourly
  - 2 meter air temperate (t2m)
  - 2 meter dew point temperature
  - surface pressure (not using today)

- Set extent to Ghana
  - Lat: -3.75 – 1.5
  - Long: 4 – 11.5

- Format NetCDF

- API for bulk download available

https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=download

Thank you!

cascade.tuholske1@montana.edu

…next time: climate projections + Python data structures

Google COLAB Tutorial: https://drive.google.com/drive/folders/15UMd-FTgG8wYqeeaseFRPlBK51Bj0qje

GitHub Tutorial: https://github.com/EqiLuo/Climate-Health-Data-Science-Workshop-COLAB

Data Links & References

Data

- ERA5: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=download
- CHIRTS: https://chc.ucsb.edu/data/chirtsdaily
- CHC-CMIP6: https://chc.ucsb.edu/data/chc-cmip6
- NASA NEX GDDP: https://www.nccs.nasa.gov/services/data-collections/land-based-products/nex-gddp-cmip6

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